

# Riphah International University Gulberg Green



**ASSIGNMENT NO : 02**

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**Department : BSSE-3**

**Course : Data Structure**

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**Date : 24/02/2026**

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### Question NO 1:

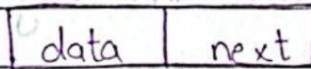
A singly linked list contains nodes where each node has two parts:

Data and Next:

Explain the following elements of a singly linked list:

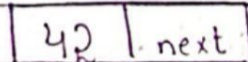
#### Node:

A node is the basic building block of a linked list. It is an object that holds two things: the data and a reference (pointer) to the next node.



#### Data field:

The data field stores the actual value or information that the node carries (eg: integer, string or any data type).



data field →

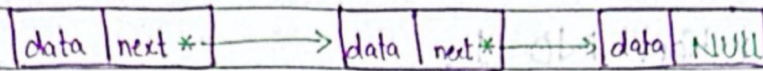
#### Next pointer:

The next pointer is a reference that points to the next node in the sequence, linking nodes together. The last node's next pointer is usually NULL (indicating the end of the list).



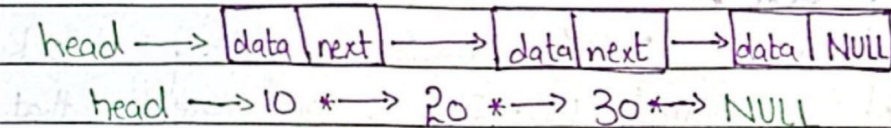
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### Head pointer:

The head pointer is a special reference that points to the first node of the list, giving access to the entire list. If the list is empty, the head is NULL.



### Part A: Insertion Operations

Write pseudocode and dry run:

#### 1) Insertion at the Beginning:

Insert 10 into list: 20  $\rightarrow$  30  $\rightarrow$  40

head  $\rightarrow$  20  $\rightarrow$  30  $\rightarrow$  40  $\rightarrow$  NULL

#### Pseudocode:

Node \*newNode;

newNode = new Node;

newNode  $\rightarrow$  data = 10;

newNode  $\rightarrow$  next = head;

head = newNode;

#### Dry Run:

Initial List

head  $\rightarrow$  20  $\rightarrow$  30  $\rightarrow$  40  $\rightarrow$  NULL

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**Step 1:**

Create new node

$\text{newNode} \rightarrow 10 \rightarrow \text{NULL}$

**Step 2:**

$\text{newNode} \rightarrow \text{next} = \text{head}$

List becomes:

$10 \rightarrow 20 \rightarrow 30 \rightarrow 40$

**Step 3:**

$\text{head} = \text{newNode}$

**Final List:**

$\text{head} \rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

**2) Insertion at End:**

Insert 50 into list  $10 \rightarrow 20 \rightarrow 30$

$\text{head} \rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$

**Pseudocode:**

Node \*ptr;

Node \*newNode;

$\text{newNode} = \text{new Node};$

$\text{newNode} \rightarrow \text{data} = 50;$

$\text{newNode} \rightarrow \text{next} = \text{NULL};$

$\text{ptr} = \text{head};$

$\text{while} (\text{ptr} \rightarrow \text{next} \neq \text{NULL}) \{$



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```
ptr = ptr → next;  
}  
ptr → next = newNode;
```

**Dry Run:**

Initial list

head → 10 → 20 → 30 → NULL

**Pointer Movement:**

ptr → 10

ptr → 20

ptr → 30

**Link new node:**

30 → 50 → NULL

**Final list:**

head → 10 → 20 → 30 → 50 → NULL

**3) Insertion at a Specific Position:**

Insert 25 at position 3 into list: 10 → 20 → 30 → 40

head → 10 → 20 → 30 → 40 → NULL

**Pseudo code:**

Node \*ptr;

Node \*newNode;

newNode = new Node;

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newNode → data = 25;

ptr = head;

```
for (int i = 1; i < position; i++) {  
    ptr = ptr → next;  
}
```

newNode → next = ptr → next;

ptr → next = newNode;

**Dry Run:**

**Initial:**

head → 10 → 20 → 30 → 40 → NULL

**Move ptr:**

ptr → 10

ptr → 20

**Linking:**

newNode → next = 30

20 → next = 25

**Final list:**

head → 10 → 20 → 25 → 30 → 40 → NULL





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## Part B: Deletion Operations

Write pseudocode and dry run:

### 4) Deletion from Beginning:

Delete first node from list:  $10 \rightarrow 20 \rightarrow 30$

head  $\rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$

#### Pseudocode:

Node \*ptr;

ptr = head;

head = head  $\rightarrow$  next;

delete ptr;

#### Dry Run:

Initial:

head  $\rightarrow 10 \rightarrow 20 \rightarrow 30$

#### Step 1:

ptr = 10

#### Step 2:

head = 20

#### Step 3:

delete ptr = 10.

#### Final List:

head  $\rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$



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### 5) Deletion from End:

Delete last node from list :  $10 \rightarrow 20 \rightarrow 30 \rightarrow 40$

head  $\rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

#### Pseudocode:

Node \*ptr;   
 Node \*predPtr;

ptr = head;  
predPtr = NULL;

```
while (ptr -> next != NULL) {  
    predPtr = ptr;  
    ptr = ptr -> next;  
}
```

predPtr -> next = NULL;  
delete ptr;

#### Dry Run:

Initial:

head  $\rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

#### Pointer movement,

predPtr = 10 , ptr = 20

predPtr = 20 , ptr = 30

predPtr = 30 , ptr = 40

#### Delete :

delete ptr = 40



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Final list:

head  $\rightarrow$  10  $\rightarrow$  20  $\rightarrow$  30  $\rightarrow$  NULL

6) Delete at Specific Position:

Delete node at position 2 from

list: 10  $\rightarrow$  20  $\rightarrow$  30  $\rightarrow$  40

head  $\rightarrow$  10  $\rightarrow$  20  $\rightarrow$  30  $\rightarrow$  40  $\rightarrow$  NULL

Pseudocode:

Node \*ptr;

Node \*predPtr;

ptr = head;

predPtr = NULL;

for(int i = 1; i < position; i++) {

    predPtr = ptr;

    ptr = ptr  $\rightarrow$  next;

}

predPtr  $\rightarrow$  next = ptr  $\rightarrow$  next;

delete ptr;

Dry Run:

Initial:

head  $\rightarrow$  10  $\rightarrow$  20  $\rightarrow$  30  $\rightarrow$  40  $\rightarrow$  NULL

ptr = 10

predPtr = NULL

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Loop Runs Once:

predPtr = 10

ptr = 20

Change Link:

10 → next = 30

list becomes:

10 → 30 → 40

Delete Node:

delete 20.

Final list:

head → 10 → 30 → 40 → NULL